Christopher Xiang

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EXPERIENCE

Hyundai America Technical Center, Inc **January 2026 - Present**
Machine Learning Engineer / Software Engineer *Ann Arbor, MI*

- Built ML surrogate models for CFD simulations, battery aging, and automotive bushing analysis, reducing analysis time from hours to near real-time inference
- Created tools to monitor system performance and assist in troubleshooting deployment issues
- Developed AI-powered plotting and visualization tools for engineering and simulation data analysis
- Built and maintained a group-wide RAG system with document ingestion pipelines
- Constructed and maintained **GitLab** CI/CD pipelines to automate testing, container builds, and deployment workflows, improving deployment reliability and reducing manual release effort

Xbox Game Camp **July 2025 - November 2025**
Gameplay Programmer / Game Designer *Remote*

- Designed and developed a procedurally generated golf roguelike with **Unity** and **C#**
- Designed and implemented core gameplay systems for spellcasting and in-game shops
- Collaborated with a five-person team using **Git** for version control and collaborative development

WolverineSoft **June 2025 - September 2025**
Gameplay Programmer *Remote*

- Developed and implemented UI elements using **Unity** and **C#**.
- Implemented controller support to improve accessibility and expand input compatibility.
- Collaborated with a small development team using Agile practices to develop and iterate on game features.
- Identified, debugged, and resolved software bugs throughout development.
- Used **Git** for version control and team collaboration.

University of Michigan Engineering | CAEN **June 2022 - May 2025**
Lab Inspector *Ann Arbor, MI*

- Onboarded and trained new team members on lab procedures, equipment handling, and maintenance best practices to ensure efficient operations
- Installed and configured over 300 new PCs and verified operation of over 500 individual computer stations
- Optimized user experience by performing preventive maintenance on over 20 computer labs and classrooms, including cleaning PCs, managing cables, and maintaining furniture arrangements

PROJECTS

X Engine **January 2025 - Present**

- Developed an integrated 2D game engine in **C++** over 9 weeks utilizing **Test Driven Development**
- Identified specification gaps, leading to improved integration tests and enhanced test case coverage
- Optimized engine performance using Visual Studio Profiler to identify and address bottlenecks
- Integrated key libraries including SDL (rendering), Box2D (physics), and Lua (scripting)

Windy Wizard **October 2024 - December 2024**

- Developed a puzzle-action platformer in **Unity**, scripting core gameplay systems in **C#**
- Leveraged Jira for project management and task coordination with a four-member team
- Achieved **3rd** place out of 15 groups during games showcase
- Edited and produced a gameplay trailer using Adobe tools

Insta485 **January 2024 - February 2024**

- Developed a full-stack image hosting web application modeled after Instagram
- Built the frontend using React, and implemented backend services with Flask and SQL
- Deployed the application on Amazon Web Services (AWS), leveraging EC2 services for hosting and storage

Network Attack Detection **August 2023 - October 2023**

- Preprocessed network traffic data using pandas, including feature selection and normalization
- Trained and evaluated multiple machine learning models using scikit-learn, such as Random Forest, SVM, and K-Nearest Neighbors, to classify network attacks
- Achieved high detection accuracy by tuning hyperparameters and applying cross-validation techniques
- Generated classification reports and confusion matrices to evaluate model performance and improve threat detection

TECHNICAL SKILLS

Coding: Python, C++, C, MySQL, PostgreSQL, Java, JavaScript, Bash, YAML, JSON

AI / ML: LLM fundamentals, vLLM, Tensorflow, Pytorch, Keras, scikit-learn, data preprocessing, deep learning
DevOps: Docker, Podman, CI/CD fundamentals, environment management, Agile, deployment troubleshooting
Cloud & Tools: AWS, Git, Jira, Visual Studio, Figma, Jupyter Notebook, Adobe Creative Suite

EDUCATION

University of Michigan, Ann Arbor | Bachelor of Science in **Computer Science**, GPA: 3.49